

Successful Bacteriophage Treatment of a Recalcitrant Intra-Abdominal Infection Caused by Multidrug-Resistant *Pseudomonas aeruginosa* in a 2-Year-Old Child

Sanchi Malhotra, MD¹, Keiko C. Salazar, PhD², Neema Pithia, MD¹, Ishminder Kaur, MD¹, Kristina Adachi, MD¹, Nanda Ramachandar, MD^{3,4}, Austin L. Terwilliger, PhD², Anthony Maresso, PhD², Robert S. Venick, MD⁵, Suzanne V. McDiarmid, MD⁵, and John S. Bradley, MD³

Antimicrobial resistance is life-threatening to pediatric patients with medical complexity who receive multiple courses of broad-spectrum antibiotics. Bacteriophages offer a safe treatment alternative when our antibiotic armamentarium is no longer sufficient. We describe successful use of bacteriophage therapy on a patient with a recalcitrant *Pseudomonas aeruginosa* infection after receiving a multiorgan transplant. (*J Pediatr* 2026;291:115005).

Antimicrobial resistance is a rising global threat and increasingly affecting pediatric populations. Pediatric patients undergoing solid-organ transplantation may harbor a microbiome that demonstrates profound antibiotic resistance caused by multiple courses of antimicrobial therapy for repeated infections and nosocomial pathogen colonization, potentially culminating in infections that are untreatable with our current antibiotic armamentarium.

Bacteriophages are ubiquitous, naturally occurring viruses with bacteria as their natural hosts that can be harnessed to cause bacteriolysis of targeted bacterial populations without effect on human cells.¹ Bacteriophages have been tried for the treatment of human infections since they were named in 1917; however, they fell out of favor when antibiotics were developed as a result of bacteriophages' narrow spectrum of activity, difficulty of mass production, and need for personalization.² Outside of a trial for dysentery caused by *Escherichia coli*, the use of bacteriophages in pediatric patients is limited to case reports, most often in patients with multidrug-resistant (MDR) infections, including those with cystic fibrosis and chronic pulmonary infection.^{1,3} Few infants have been reported to receive phages, with 2 successful outcomes for infants with biliary atresia after liver transplant; one with vancomycin-resistant *Enterococcus faecium* infection and another with MDR *Pseudomonas aeruginosa* infection.⁴⁻⁷ Here, we present a case of a persistent, uncontrolled MDR *P aeruginosa* intra-abdominal infection in an immunosuppressed 2-year-old multiorgan transplant recipient that was treated successfully with bacteriophages after failure of traditional antibiotic therapy.

Case

A 2-year-old female patient with a history of biliary atresia underwent Kasai procedure at 4 months of age at a hospital in Mexico. In the months after the procedure, her course was complicated by volvulus requiring bowel resection (with ileostomy status after take-down), causing short bowel syndrome with enterocutaneous fistula, intestinal failure, liver disease with portal hypertension, multiple central line-associated bloodstream infections, and recurrent cholangitis. Because of intestinal failure-associated liver disease, she was referred to the University of California, Los Angeles, for multiorgan transplant, which she underwent at 2 years of age including whole liver, duodenum, pancreas, jejunum, ileum, and right-transverse colon. She received piperacillin/tazobactam, fluconazole, and ganciclovir prophylactically after transplantation.

She required reoperation on postoperative day (POD) 2 for evolving abdominal compartment syndrome with decompressive laparotomy, gastrointestinal reconstruction with a colostomy of the transplanted bowel, and temporary abdominal wall closure. On POD 5, she became febrile, and her antimicrobial therapy was changed to meropenem, vancomycin, and micafungin empirically (Figure 1). Results of blood culture tests were negative. Her fever improved; however, on POD 12, she was found to have a microperforation at her jejunal site of anastomosis, which required repair. Postoperatively an enterocutaneous fistula persisted. She had recurrence of fever on POD 18, and a

MDR	Multidrug-resistant
POD	Postoperative day

From the ¹Division of Infectious Diseases, Department of Pediatrics, University of California Los Angeles, Los Angeles, CA; ²TAILOR Labs, Department of Molecular Virology and Microbiology, Baylor College of Medicine, Houston, TX; ³Division of Infectious Diseases, Department of Pediatrics, University of California San Diego, Rady Children's Hospital San Diego, San Diego, CA; ⁴Department of Pediatrics, Naval Medical Center San Diego, San Diego, CA; and ⁵Division of Gastroenterology, Department of Pediatrics, University of California Los Angeles, Los Angeles, CA

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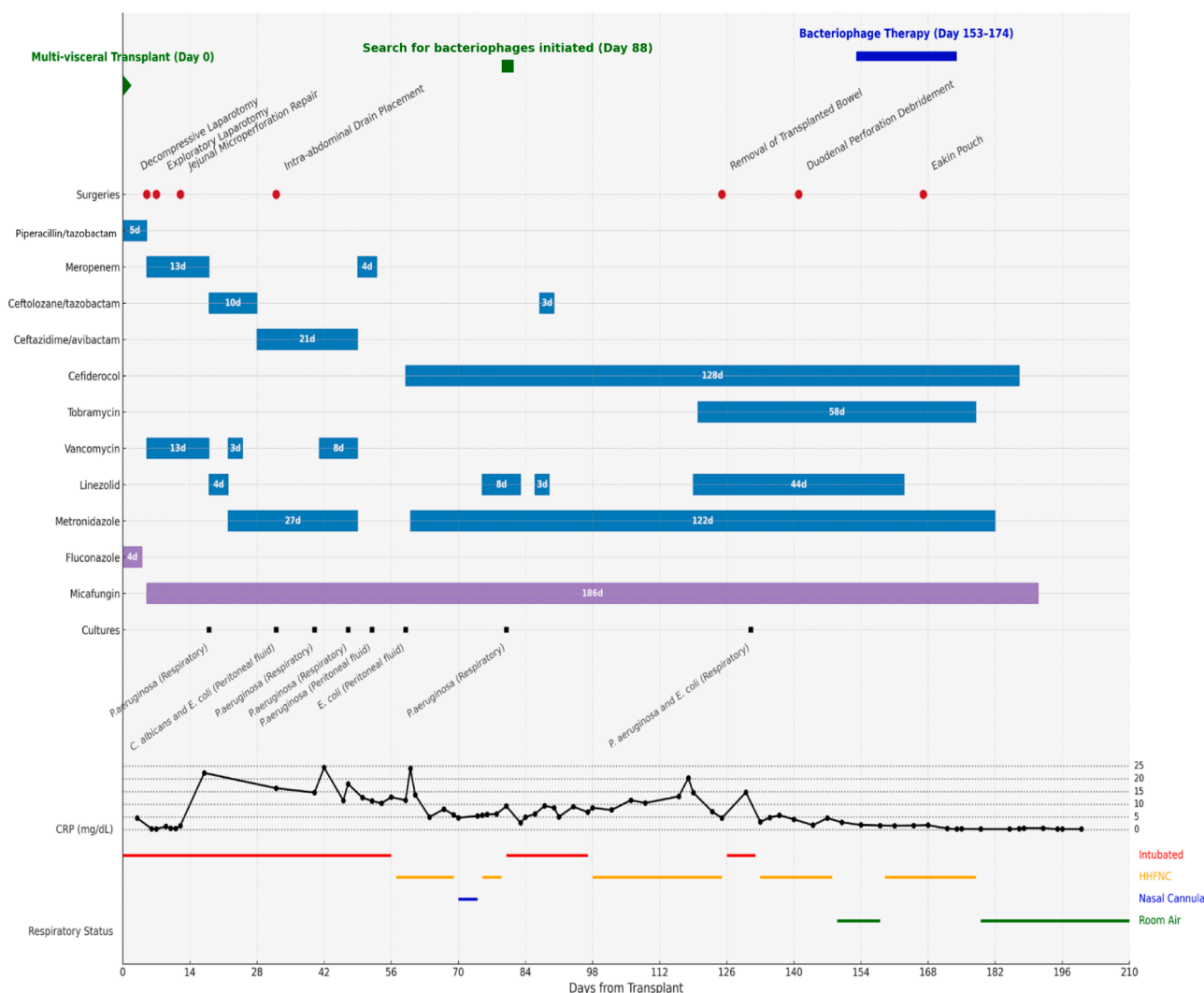


Figure 1. Key events during patient hospitalization are shown in this timeline, including surgeries, antimicrobials, cultures, C-reactive protein (CRP), and respiratory status to demonstrate the patient course leading up to a search and administration of bacteriophages. HHFNC, heated high-flow nasal cannula.

respiratory culture obtained from the endotracheal tube yielded growth of a moderate amount of carbapenem-resistant *P aeruginosa*, which tested negative for the presence of KPC, NDM, IMP, VIM, and OXA-48-like carbapenemases (Table I). Thus, she was transitioned from meropenem to ceftolozane-tazobactam (Figure 1).

On POD 24, she underwent an intestinal biopsy, which showed severe T-cell-mediated rejection; therefore, she was treated with pulse-dose corticosteroids and 5 courses of antithymocyte globulin. During this time, she clinically worsened acutely, and conventional microbiologic studies were unrevealing. A plasma cell-free DNA test was sent, resulting with >210 000 molecules of *E coli* per 100 nanoliters (positive for CTX-M and negative for KPC) without *P aeruginosa*; antibiotics were empirically changed on POD 29 to ceftazidime-avibactam and metronidazole (Figure 1). As a result of

worsening abdominal distention on POD 33, a computed tomography scan was obtained, which showed a large loculated peritoneal fluid collection. She underwent drainage and drain placement, with removal of 1800 mL of fluid; fluid cultures yielded growth of carbapenem-resistant *E coli* and antifungal drug-susceptible *Candida albicans*. Repeat peritoneal fluid cultures on POD 48 grew MDR *P aeruginosa*. She continued to have a labile clinical course during this time, with need for intermittent ventilatory support and alternating care between the step-down intensive care units. Multiple respiratory cultures obtained in the 12 weeks after her transplant documented persistence of *P aeruginosa*. By POD 56, she was persistently febrile and clinically labile, with *P aeruginosa* isolates with increasing resistance (Table I); gram-negative bacillary coverage was changed to cefiderocol. On POD 76,

Table 1. Antibiotic susceptibility testing for *Pseudomonas aeruginosa* isolates

Antibiotics	Minimum inhibitory concentration								
	POD 18*	POD36*	POD 43*	POD 48†	POD 56†	POD 76*	POD81*	POD 87*	POD 130*
Cefepime	16 (I)	16 (I)	32 (R)	>32 (R)	4 (S)	>32 (R)	>32 (R)	>32 (R)	>32 (R)
Ceftazidime	8 (S)	16 (I)	>32 (R)	32 (R)	4 (S)	>32 (R)	>32 (R)	>32 (R)	>32 (R)
Ceftazidime/avibactam	8 (S)	8 (S)	32 (R)	32 (R)	4 (S)	32 (R)	32 (R)	32 (R)	32 (R)
Ceftolozane/tazobactam	2 (S)	2 (S)	4 (S)	8 (I)	1 (S)	4 (S)	8 (I)	8 (I)	>32 (R)
Cefiderocol		2 (S)	2 (S)	2 (S)	2 (S)	I	I	I	I
Imipenem	>8 (R)	>8 (R)	>8 (R)	>8 (R)	>8 (R)	>8 (R)	>8 (R)	>8 (R)	>8 (R)
Meropenem	>16 (R)	>16 (R)	>16 (R)	>16 (R)	8 (R)	>16 (R)	>16 (R)	>16 (R)	>16 (R)
Ciprofloxacin	0.5 (S)	0.5 (S)	1 (I)	≤0.25 (S)	0.5 (S)	0.5 (S)	0.5 (S)	0.5 (S)	1 (I)
Levofloxacin	2 (I)	2 (I)	4 (R)	2 (I)	>1 (S)	4 (R)	2 (I)	4 (R)	4 (R)
Piperacillin + tazobactam	16 (S)	16 (S)	128 (R)	>128 (R)	16 (S)	128 (R)	128 (R)	128 (R)	128 (R)
Tobramycin	≤1 (S)	≤1 (S)	≤1 (S)	≤1 (S)	≤1 (S)	≤1 (S)	≤1 (S)	≤1 (S)	>16 (R)
Imipenem/relebactam		0.12/4 (S)	2/4 (S)	R	0.12/4 (S)				

I, intermediate; POD, postoperative day, from multivisceral transplant; R, resistant; S, susceptible.

*Isolated from respiratory culture.

†Isolated from abdominal fluid culture.

the *P. aeruginosa* isolate recovered had only intermediate susceptibility to cefiderocol, prompting addition of intravenous tobramycin. On POD 125 from transplant, the patient underwent exploratory laparotomy and required subtotal allograft enterectomy (STE) because of concerns for ongoing severe rejection.

Search for Bacteriophage Therapy

Given ongoing fevers, intra-abdominal leakage, and increasingly resistant *P. aeruginosa* in abdominal fluid and respiratory cultures, a search for bacteriophage therapy was initiated 12 weeks after her initial multiorgan transplant (Figure 1). The patient's *P. aeruginosa* isolates were sent sequentially to 3 research laboratories known to have a library of *Pseudomonas* bacteriophages. The first 2 laboratories did not have active bacteriophages; however, after 9 weeks of searching, TAILOR labs at Baylor College of Medicine identified 2 bacteriophages with activity against her clinical isolates. *E. coli* phages were considered but not pursued because her isolates retained susceptibility to antibiotic options.

Given the patient's ongoing critical illness, anti-*Pseudomonas* bacteriophage therapy with an active 2-phage cocktail was initiated 4 weeks after STE, with parental consent, institutional review board approval, and emergency-use Food and Drug Administration authorization. Therapy was planned for 28 days, during which she remained on the same antimicrobials (Figure 1).

Clinical Response to Bacteriophage Therapy

The bacteriophages were infused intravenously at a dose of 1×10^9 PFU/mL (1 mL) every 12 hours, which was not associated with any adverse event. However, a profound elevation in lactate was noted 2 days after commencing phage therapy (95 mg/dL to 169 mg/dL), which ultimately was thought to be secondary to prolonged linezolid use, which was discontinued on day 10 of bacteriophage therapy. Her lactate slowly normalized and remained low thereafter. On day 11 of phage therapy, findings of a computed tomography scan of the

abdomen showed an abscess at her previous duodenostomy tube site. The abscess could not be drained because of dense fibrosis preventing a safe approach, so medical management alone was continued. Phage neutralization assays were collected during the first 2 weeks of treatment (see the Methods section); results were not available during the treatment period, but ultimately showed a significant decrease in phage titers in the neutralization assay against the 2-phage cocktail by day 14 of bacteriophage therapy (Figure 2).

Approximately 6 weeks after STE and 15 days after commencing bacteriophage therapy, our patient began to have steady clinical improvement. She was transferred out

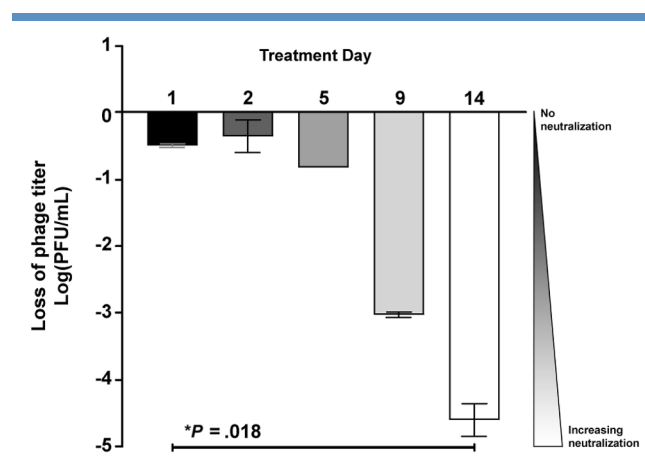


Figure 2. Neutralization of *Pseudomonas* bacteriophage cocktail from patient serum samples collected on treatment days 1–14. Shown is the decrease in phage titer after incubation of the 2-phage cocktail in the bacterial host, in the presence of patient serum collected 30 minutes prephage and 5–120 minutes postphage administration over the course of treatment. Bars indicate the average of 3 technical replicates, with error bars showing SD. Significance was determined using the Mann-Whitney *U* test for nonparametric datasets. PFU, plaque-forming units.

of the intensive care unit and remained stable in the observation unit in room air and without need for continuous renal-replacement therapy or vasopressor support. Given her clinical response, resolution of fevers, and normalization of C-reactive protein, the bacteriophage therapy was discontinued after a 21-day course. There was not concern for ongoing infection, so antimicrobial agents were tapered sequentially every 72 hours and then stopped for the first time after 6 months of continuous therapy posttransplant. Over the 6 months after completing bacteriophage therapy, she remains hospitalized, has not received an antibiotic, remains clinically stable, and awaits a repeat multiorgan transplant.

Methods

All 4 strains of *P aeruginosa* collected from the patient over the course of 2 months were found to be susceptible to a bacteriophage cocktail containing phages HC18 and HC20 (Supplemental Table 1, available at www.jpeds.com). The phages were previously identified and isolated from Houston wastewater samples. The following steps are described in detail in Green et al.⁸

Phage Purification

The phages were replicated at TAILΦR Labs at Baylor College of Medicine in a *P aeruginosa* isolate called PA01.⁹⁻¹¹ The phages were subsequently purified using salt precipitation and ultracentrifugation with a cesium chloride gradient followed by dialysis.

Cocktail Formulation

The following steps were performed in a class II, type A2 biosafety cabinet in a bacteria-free space. On the basis of predicted titer, the 2 phages were combined and run through an Endotrap HD (Ilex Life Sciences; catalog number LET0012) column to reduce endotoxin. As a result of toxicity limits and the small size of the patient, the cocktail was run through an additional Endotrap HD column to remove residual endotoxin. The cocktail was then filtered to create a sterile preparation for clinical use, using a 0.22-μm polyvinylidene difluoride syringe filter (MilliporeSigma; catalog number SLGVR33RS) and aliquoted into sterile glass vials. Samples were collected from each vial for USP<71> sterility testing, performed by Nova Biologicals (<https://www.novatx.com/>). An additional sample was sent for acid-fast bacteria testing to detect fastidious organisms.

Safety and Efficacy Testing

In addition to sterility testing, samples were sent to Nova Biologicals for USP<85> endotoxin quantification. *Pseudomonas* exotoxin A was quantified using a sandwich enzyme-linked immunosorbent assay kit (Biotang; catalog number HU9457). Before formulation, the phages were verified and checked for virulence factors, toxins, and integrases using whole-genome sequencing (Supplemental Table 2, available at www.jpeds.com).

Cocktail efficacy was tested by double agar overlay titer against all 4 isolates.¹² This same method was used to determine cocktail stability over 1 week. A sample of cocktail was diluted in saline and stored at the approximate pharmacy formulation with doses prepared for 1 week at a time.

Patient Phage Neutralization Assay

For the patient phage neutralization assay,¹³ blood samples were drawn from the patient before (30 minutes) and after (5, 30, 60, or 120 minutes) phage administration on treatment days 1, 2, 5, 9, and 14. The samples were centrifuged at $1000 \times g$ for 10 minutes to separate and collect the serum; 1 μL of phage cocktail was incubated in 100 μL of serum at 37°C, shaking, for 30 minutes. The phage-serum mixtures were then titrated by double agar overlay on the target bacterial isolate. The same process was performed using phage buffer as a negative control. Each serum sample was tested in triplicate. Titers were transformed by log₁₀, then data-points were normalized by subtraction from the average control cocktail titer. Statistics were calculated and graph was generated using GraphPad Prism, version 10.5.0 (Figure 1).

Discussion

Our patient developed an extensively MDR *P aeruginosa* postoperative infection that persisted despite antibiotic therapy and multiple surgical debridements over 6 months. Ultimately, she had sustained improvement after bacteriophage therapy, without need for further antimicrobial therapy or surgery. She remains at high risk for recurrent MDR infections after a repeat transplant and may need new phages against updated isolates as her bacterial flora may maintain resistance in response to phage therapy.

Finding bacteriophages for treatment of an acutely ill patient is a challenging and time-consuming process. A patient's bacterial pathogen (including strains with different colony morphologies) must be sent to a research laboratory with a library of phages specific for the patient's pathogen for susceptibility testing to assess whether any strains in the library are active against the patient's isolate. The search for phages may require testing for susceptibility against multiple institutions' libraries. It took 9 weeks from when we first sent her isolates to multiple laboratories to receive a sterile, active phage product for infusion from the third laboratory we approached (TAILΦR Laboratory at Baylor University). We were fortunate because this phage cocktail had already been prepared and tested and was in storage. However, more commonly, several more weeks would have been needed for cocktail preparation, sterility, and endotoxin testing, during which our acutely ill patient would have remained tenuous clinically. Procuring bacteriophages and gaining regulatory approval may take even longer for stable, nonemergent, chronic infections such as are characteristic of cystic fibrosis.^{14,15}

Intravenous and oral bacteriophage therapy has been used successfully in vivo for intra-abdominal infections in animal models,¹⁶⁻¹⁸ and the aforementioned infant with

vancomycin-resistant *Enterococci* infection.⁵ Many concerns exist for the ability of phages to reach a susceptible intestinal bacterial host after oral administration and passage through the acidic gut environment. For intra-abdominal infections, loculated collections arising from infection and inflammation may present physical barriers that limit access of phage to the pathogen. Moreover, some bacteria may have increased phage resistance through a variety of cellular mechanisms after exposure to treatment phages (as demonstrated in our case), or from previous exposure to other commensal phages.¹⁸ However, bacteriophages appear uniquely capable of penetrating tissues to reach their bacterial targets, and even a few phages entering an abscess may multiply locally within bacterial hosts to sustain antibacterial activity and substantially reduce or eradicate the pathogen. In addition, synergy may be present with some antibiotic/phage combinations against pathogens, including *P aeruginosa*.^{19,20} In some cases, bacteriophage exposure has been shown to induce bacterial changes that paradoxically restore susceptibility to antibiotics for which the bacteria previously had been resistant.²¹

Emergence of phage-resistant bacterial variants during treatment, and serum neutralization of phages through anti-phage antibodies can also limit efficacy. In our patient we saw her phage titer in vitro decrease significantly by day 14 of therapy, likely as the result of neutralization by an increase in host antibodies, although the patient continued to improve clinically. Because of insufficient blood sample volume from the patient, we were not able to test neutralization against the phages individually, nor against a nonspecific control phage. This limits our ability to conclude whether neutralization was due to antibodies specifically against 1 of the 2 strains, or if unrelated to antiphage antibodies. It is possible that the additional therapy administered beyond day 14 was not efficacious; however, data were not available in real time, and it is difficult to predict at what time point phage neutralization is a significant concern. Although concurrent antibiotic and surgical therapies limit our ability to attribute success solely to bacteriophage therapy, the patient had previously been unable to have antibiotics discontinued without clinical decompensation. After bacteriophage therapy, she has shown no evidence of recurrent infection and has required no further debridement nor antibiotics.

Prospective clinical trials for phage therapy in adults are currently underway for a variety of sites of infection and pathogens.²²⁻²⁴ Pediatric trials for phage therapy of tonsillitis are active in Uzbekistan,²⁵ but overall research efforts to study phage therapy in critically ill pediatric populations are lacking. Challenges in identification of active bacteriophage cocktails, the technical difficulty of purification and preparation of bacteriophages for clinical use, and regulatory hurdles further hamper the application of this novel therapeutic approach for salvage therapy, and make prospective, controlled clinical trials difficult in pediatrics.²⁶

Bacteriophages represent a safe therapeutic option when antimicrobial resistance has led to an untreatable infection,

with potential for eradicating deep seated infections in high-risk patients. A concerted, multipronged and systematic approach to study and define appropriate use of bacteriophages in children with severe infections is needed to meet the dire threat of growing antimicrobial resistance. Informed consent was obtained from the patient's parents in accordance with ethical guidelines for pediatric research. ■

CRediT authorship contribution statement

Sanchi Malhotra: Writing – review & editing, Writing – original draft, Conceptualization. **Keiko C. Salazar:** Writing – original draft, Methodology, Formal analysis, Data curation. **Neema Pithia:** Writing – original draft, Data curation. **Ish-minder Kaur:** Writing – review & editing, Project administration. **Kristina Adachi:** Writing – review & editing. **Nanda Ramachandar:** Writing – review & editing, Supervision, Conceptualization. **Austin L. Terwilliger:** Writing – review & editing, Supervision, Formal analysis, Data curation. **Anthony Maresso:** Supervision, Methodology, Formal analysis, Data curation. **Robert S. Venick:** Supervision, Conceptualization. **Suzanne V. McDiarmid:** Supervision, Conceptualization. **John S. Bradley:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of Competing Interest

A.T. and A.M. disclose restricted stock in their co-founded company, Phiogen. The remaining authors do not have any additional disclosures. The views expressed in this article are those of the author(s) and do not necessarily reflect the official policy or position of the Department of the Navy, Department of Defense, or the US Government.

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Reprint requests: Sanchi Malhotra, MD, 10833 Le Conte Ave, 22-340 MDCC, Los Angeles, CA 90095. E-mail: smalhotra@mednet.ucla.edu

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